

Integration of the FINS Protocol into the Open-Source HMI System FUXA

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- Introduction
- Related Works
- Methodology
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- Conclusion and Future Work

- Omron PLCs dominate production lines at PT XYZ ($> 90\%$ machines).
- Factory Interface Network Service (FINS) is most used in omron PLC.
- Commercial HMI/SCADA systems are costly and vendor-locked.
- Open-source HMI FUXA lacks native FINS support.

Introduction – Problem Statement

- Lack of native FINS support prevents FUXA adoption in Omron-centric plants.
- Middleware-based integration increases latency and complexity.
- Alarm reliability and real-time visualization are critical requirements.

RQ1

How can FINS protocol support be implemented in FUXA with reliability and feature parity comparable to built-in protocols?

RQ2

How can stable, low-latency read/write operations be achieved for real-time visualization?

Introduction – Main Contributions

- First native integration of Omron FINS into the FUXA HMI platform.
- Backend driver (`DeviceFins`) and frontend configuration support.
- Empirical evaluation of latency, reliability, resource usage, and alarms.
- Identification of alarm latency bottleneck in polling-based HMI systems.

Related Works – Systematic Literature Review I

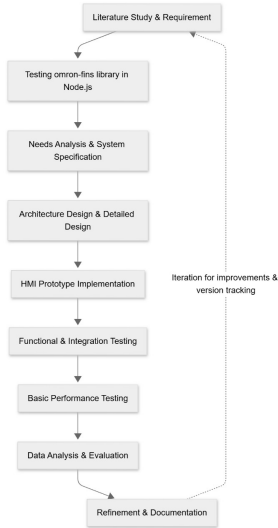
Author & Year	Platform	Domain	Protocol	Main Contribution
Uddin et al. (2022)	Node-RED, Grafana	Solar RO plant	HTTP, MQTT	Open-source SCADA feasibility
Omidi et al. (2023)	Node-RED SCADA	Renewable energy	MQTT, REST	Modular real-time SCADA
Almas et al. (2014)	Open SCADA	Smart grid	DNP3	Polling performance analysis
Hazdi et al. (2017)	Web-based SCADA	Home automation	OPC	PLC-based web SCADA
Arnoud et al. (2021)	FUXA	Cybersecurity testbed	Modbus TCP	FUXA as visualization layer
Salazar et al. (2024)	ICSNet + FUXA	Industrial honeynet	Multi-protocol	ICS security visualization
Gil et al. (2022)	Framework study	Industrial systems	Multi-protocol	Interoperability analysis
Giehl et al. (2024)	OpenPLC	Industrial control	OPC UA	Secure semantic integration
This work	FUXA	Manufacturing	Omron FINS	First native FINS integration

Related Works – Research Gap

- Existing studies focus on Modbus, OPC UA, MQTT.
- No native open-source HMI support for Omron FINS.

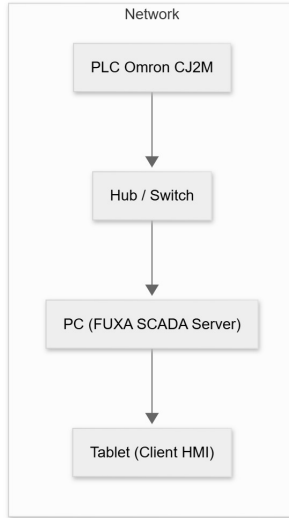
- Iterative development approach:
 - Requirements analysis
 - System design
 - Implementation
 - Evaluation

Methodology – Research Workflow



- Omron PLC layer
- Node.js backend (FUXA + DeviceFins)
- Angular-based HMI frontend

Methodology – Architecture Diagram



- Node.js DeviceFins backend driver
- FINS UDP and TCP communication
- PLC memory access (DM, CIO, W)
- Auto-reconnect and timeout handling

- Device configuration interface
- Tag editor for FINS memory areas
- Event-driven real-time UI updates

Methodology – Evaluation Metrics and Justification

- System performance evaluated using five key metrics reflecting real-time HMI–PLC requirements.
- KPI thresholds derived from conventional industrial HMI operation at PT XYZ.

Metric	KPI Target
Read/write latency	≤ 200 ms
Success rate	$\geq 95\%$
CPU / Memory usage	$\leq 30\%$ / ≤ 500 MB
Alarm detection time	≤ 200 ms
Startup time	≤ 10 s

- Test workload: 1000 read operations with 10 concurrent requests.
- Configuration represents typical status and alarm density in industrial HMIs.
- Metrics directly support RQ1 (functional parity) and RQ2 (real-time performance).

Results – Performance Summary

- Average read/write latency: **12 ms**
- Communication success rate: **99%**
- CPU usage: **2%**
- Memory usage: **462 MB**

Results – Evaluation Metrics

Table: Summary of Technical Evaluation Results

Metric	Target	Observed
Read/write latency (avg.)	≤ 200 ms	12 ms
Communication success rate	$\geq 95\%$	99%
Server CPU usage	$\leq 30\%$	2%
Server memory usage	≤ 500 MB	462 MB
Alarm detection time	≤ 200 ms	1000 ms (formal)
Application startup time	≤ 10 s	3.14 s

Results – FUXA Device Configuration (FINS)

Connections Property

Name

PLC_Mesin_Cutting

Type

Fins

Polling

350 ms

Enable

☒

FINS IP Address

192.168.0.1

SA1 (PC Node)

234

DA1 (PLC Node)

1

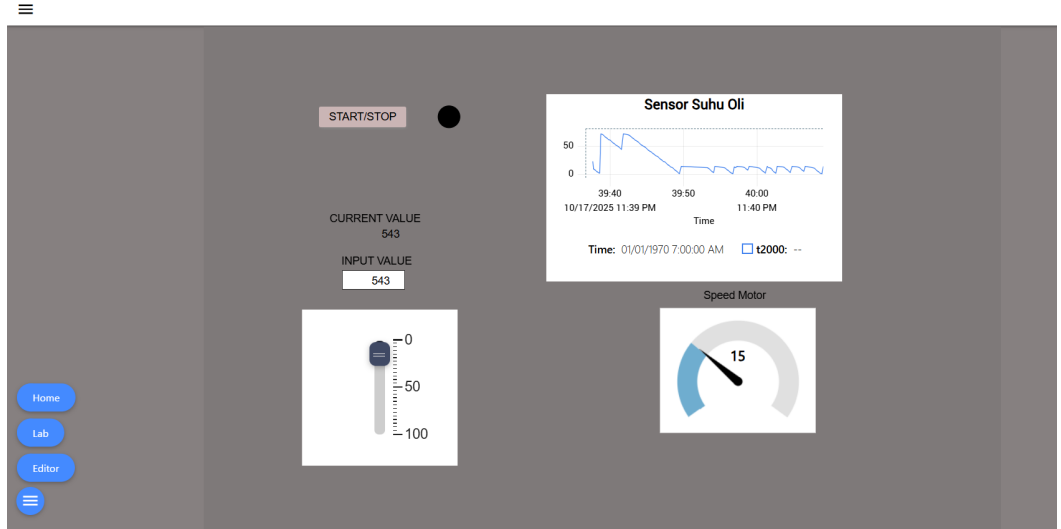
Protocol

UDP

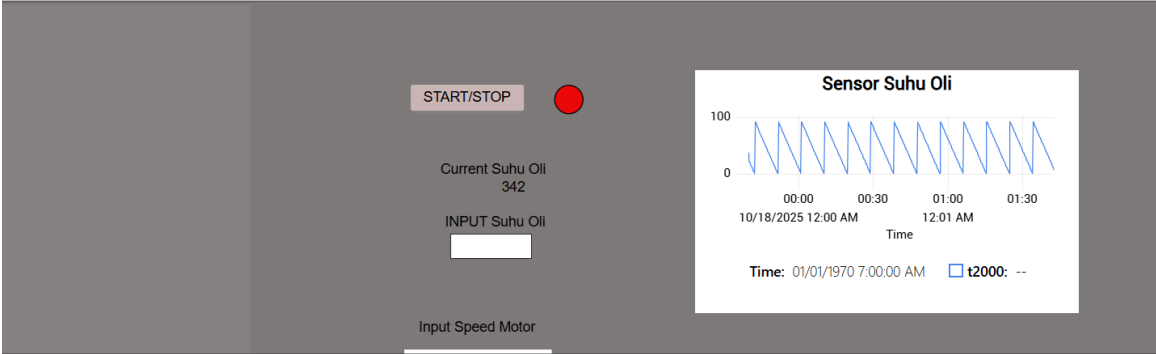
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Results – Real-Time HMI Visualization



Results – Alarm Monitoring via FINS



Alarms			
Date/Time	Text	Priority	Group
2025.10.18 00:01:34	Suhu Terlalu Panas,bahaya meledak	High High	

Results – Alarm Latency Analysis

- Observed alarm delay: **1000 ms**
- Cause: Global 1-second polling interval
- Limitation originates from HMI alarm engine

Conclusion – Research Questions Revisited

- **RQ1 Answered:** Native FINS integration was successfully implemented in FUXA with reliability and feature completeness comparable to existing protocols.
- **RQ2 Answered:** The implemented FINS connector achieved stable communication with low latency suitable for real-time HMI visualization.

Conclusion – Key Findings

- First native FINS integration for FUXA.
- Reliable low-latency communication with Omron PLCs.
- Alarm latency identified as main system limitation.

Conclusion – Future Work

- Event-driven or interrupt-based alarm mechanism
- Security enhancement (TLS, VPN isolation)
- Multi-protocol interoperability expansion

Thank You

Questions?